Financial Health Score (FHS) — Formula Derivation and Literature Basis

For Adviser Validation

Project: SmartSpend — AI-Assisted Mobile Financial Tracking and Advisory Application

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1. Overview

The SmartSpend Financial Health Score (FHS) is a **Prototype Observed Financial Health Indicator (FHI)** that computes a 0–100 score from the user's actual logged transaction data — not from self-report surveys. The FHS operates in two modes depending on whether the user has set their monthly income.

The FHS was derived by adapting the **CBA–Melbourne Institute Observed Financial Health Indicator** (Comerton-Forde et al., 2018), which uses actual bank-record data to derive a 0–100 observed scale with equal-weight components. SmartSpend adapts this transaction-based approach to a mobile SQLite database instead of bank records.

2. FHS Score Interpretation Labels

The computed FHS (0–100) maps to a qualitative label and icon displayed in the SmartSpend user interface. The bands are spaced in 10-point increments, consistent with the 0–100 observed scale of the CBA–Melbourne Institute Observed Financial Health Indicator (Comerton-Forde et al., 2018). The "Needs Work" floor (below 60) sits below the Fair band and signals that one or more score components require immediate attention.

Score Range	Label	Icon
90–100	Excellent	👑
80–89	Great	🏆
70–79	Good	⭐
60–69	Fair	🌱
Below 60	Needs Work	📉

3. The Exact Formulas

All formulas below are verified against the actual source code (`lib/services/score_service.dart`).

Full Mode (when monthly income is set)

Component 1: Savings Rate (25 pts)

```
savingsRate = (monthlyIncome − totalSpent) / monthlyIncome
score = 25 × min(1, savingsRate / 0.20)
```

- Full 25 pts when saving ≥ 20% of income
- Scales proportionally below 20%
- 0 pts when spending equals or exceeds income
- **Literature basis:** Warren & Tyagi (2005) — the 50/30/20 rule establishes 20% savings as the standard personal finance target
- **Framework alignment:** FinHealth Score® Save pillar (Financial Health Network, 2021)

No-income fallback: 20 pts if totalSpent < ₱5,000; 15 pts if < ₱10,000; 10 pts otherwise. This tiered proxy accommodates users in the informal economy who do not track formal income (UNSGSA, 2021).

Component 2: Overspend Control (25 pts)

```
dailyBudget = monthlyIncome / daysInMonth
hardOverDays = days where daily spending exceeded dailyBudget (multiple items)
softOverDays = days where a single large purchase caused the overspend
weightedOver = hardOverDays + (softOverDays × 0.5)
score = 25 × (1 − weightedOver / activeDays)
```

- Full 25 pts when no days exceeded the daily budget
- Soft overspend (one-off large purchase like tuition) counts at 0.5× penalty
- Hard overspend (habitual daily overspending) counts at 1× penalty
- **Literature basis:** Kahneman & Tversky (1979) — Prospect Theory: losses are more salient than gains; Thaler & Sunstein (2008) — Nudge Theory: framing spending as a "loss against budget" reinforces behavioral correction
- **Framework alignment:** FinHealth Score® Spend pillar — spending less than income

Component 3: Budget Adherence (25 pts)

```
score = 25 × (onBudgetCategories / totalBudgetCategories)
```

- Full 25 pts when all category budgets are on track
- 0 pts when all budgets are exceeded
- **If no budgets are set: auto-award 25 pts** (not penalized for lack of setup)
- **Literature basis:** FinHealth Score® Plan pillar (Financial Health Network, 2021) — budget adherence as a core health behavior; BSP CFIS (2025) — fewer than 50% of Filipino households maintain any

written budget, making budget-setting itself a measurable behavior

Component 4: Logging Consistency (25 pts)

```
loggedDays = number of distinct days with at least one expense entry
activeDays = days elapsed in current month (capped at daysInMonth)
score = 25 × (loggedDays / activeDays)
```

- Full 25 pts when logging every active day
- Scales proportionally with missed days
- **Literature basis:** Deci & Ryan (2000) — Self-Determination Theory: habit formation requires consistent, intrinsically motivated behavior; the Competence dimension drives long-term engagement
- **Framework alignment:** CBA–Melbourne Institute data completeness metric (Comerton-Forde et al., 2018)

Adjustments (applied after the 4 components)

Adjustment	Formula	Basis
Warning Decay	−5 pts/day, max 3 days (−15 pts total)	Kahneman & Tversky (1979) — Prospect Theory: losses create stronger behavioral responses than equivalent gains. Applied only to discretionary categories (Bills, Health, Education are exempt).
Gap Adjustment	−3 pts/day unlogged (max −15); +2 pts/day clean (max +10)	Behavioral finance consistency principle (Ariely, 2008; Duhigg, 2012). Penalty > reward follows Prospect Theory asymmetry. Capped to prevent gaming.

Lightweight Mode (when income is NOT set)

Used when the user has not configured monthly income. Redesigns 2 of 4 components to use only spending data.

Component 1: Spending Restraint (25 pts)

```
With spending limit set:
  ratio = totalSpent / spendingLimit
  score = 25 × (1 − ((ratio − 0.80) / 1.20))   [clamped 0–25]

Without spending limit:
  wantRatio = wantSpending / totalSpending
  score = 10 + 15 × (1 − ((wantRatio − 0.30) / 0.40))   [clamped 10–25]
```

- Full 25 pts at ≤80% of spending limit (grace zone)
- Scores 0 at 2× the limit
- No-limit fallback: 25 pts at ≤30% wants; 10 pts at ≥70% wants
- **Literature basis:** Kahneman & Tversky (1979) — distinguishing discretionary from necessary spending as a behavioral health signal. The 80% grace zone reflects a personal finance buffer principle.

Component 2: Logging Consistency (25 pts)

Same formula as Full Mode Component 4.

Component 3: Category Balance (25 pts)

```
discretionaryTotal = total spending excluding Bills, Health, Education
topRatio = topDiscretionaryCategory / discretionaryTotal
score = 25 × (1 − ((topRatio − 0.40) / 0.60))   [clamped 0–25]
```

- Full 25 pts when no single discretionary category exceeds 40% of discretionary spend
- Scores 0 when one category accounts for 100%
- Bills, Health, and Education are **exempt** (non-discretionary fixed obligations)
- **Literature basis:** Financial Health Network FinHealth Score® Spend pillar — spending distribution as a health indicator. BSP CFIS (2025) confirms exempt categories dominate Filipino household budgets for legitimate reasons.

Component 4: Habit Streak (25 pts)

```
currentStreak = consecutive days ending today (or yesterday) with ≥1 expense logged
score = 25 × min(1, currentStreak / 14)
```

- Full 25 pts at a 14-day consecutive logging streak
- Scales proportionally below 14 days
- **Literature basis:** Deci & Ryan (2000) SDT Competence dimension; Lally et al. (2010) — habit formation begins to anchor in the earliest phase of consistent behavior (~14 days represents initial tracking routines before deeper automaticity at ~66-day mean)

4. Why Equal Weights (25 pts each)?

Three academic justifications:

1. **CBA–Melbourne Institute precedent** — their Observed FHI uses equal-weighted transaction-derived components summing to 100 (Comerton-Forde et al., 2018)
2. **OECD/JRC Handbook on Composite Indicators** (2008) — formally establishes equal weighting as the methodologically defensible default when there is insufficient empirical basis for differential weights
3. **Transparency** — users can understand "each pillar is worth 25 points" without needing a finance degree

We do not have the multi-year longitudinal behavioral data needed to justify differential weights for this population. Equal weighting is the honest, defensible choice.

5. Alignment with Established Frameworks

SmartSpend FHS Component	CBA–Melbourne Institute (2018)	FinHealth Score® (FHN)	CFPB Financial Well-Being Scale (2017)
Savings Rate	Observed savings behavior from bank records	Save pillar: savings rate and buffer	Q7: "I could handle a major unexpected expense"
Overspend Control	Observed overspending days (weighted)	Spend pillar: spending less than income	Q2: "My finances control my life" (reverse)
Budget Adherence	Category-level spending vs. limits	Plan pillar: on track with financial goals	Q9: "My finances are on track"
Logging Consistency	Data completeness metric	(Not directly measured)	(Not directly measured)
Warning Decay	(Our addition)	(Our addition)	(Our addition)

Key distinction: The CFPB scale is self-reported (users rate their own feelings). The Melbourne Institute and FinHealth Score use transaction data like SmartSpend. The Warning Decay mechanism is our original design contribution, grounded in Prospect Theory.

6. Comparison with Other Financial Health Score Systems

To position the SmartSpend FHS within the broader landscape of financial health measurement tools, the table below compares it with established and emerging score systems. Comparison dimensions include data source, score scale, formula transparency, and the specific advantage SmartSpend offers — particularly its fully disclosed formula and offline, transaction-based computation suited to the Filipino context.

System	Organization	Scale	Data Source	Formula Disclosed	SmartSpend Advantage
FinHealth Score®	Financial Health Network (US)	0–100	8 survey indicators	Partially disclosed	Transaction-based, not survey; fully disclosed
CFPB Financial Well-Being Scale	CFPB (US)	0–100	10-question survey	Standardized scoring table	Measures behavioral outcomes, not psychological states
MindsBudget	MindsBudget (web)	0–100	Uploaded bank statements	Component structure disclosed	Offline computation; no bank API needed
Wingman Money	Wingman (AU)	0–100	Linked bank accounts	Not disclosed	Fully disclosed formula
BudgetPH / KindlyF	KindlyF (PH)	0–850	User-entered data	Not disclosed	Only direct PH competitor with behavioral score; SmartSpend formula is fully disclosed
Alkansya AI	Alkansya (PH)	0–100 (claimed)	Transactions	Not disclosed	SmartSpend's formula is fully documented; Alkansya's is undisclosed
Artha Engine	Artha (India)	0–100	Income, debt, savings, investments	Weighted formula disclosed	Broader investment orientation; SmartSpend is expense-tracking focused
SmartSpend (Full Mode)	Lucid Frame (PH)	0–100	Recorded transactions + income	Fully disclosed	Offline, dual-mode, Filipino context
SmartSpend (Lightweight Mode)	Lucid Frame (PH)	0–100	Spending behavior only	Fully disclosed	For students, irregular-income users

Transparency note: SmartSpend is the only system in this comparison whose complete scoring formula — including every calibrated constant and adjustment rule — is fully documented and reproducible from the source code (`lib/services/score_service.dart`). Competitor formulas marked "Not disclosed" are proprietary; where SmartSpend claims an advantage, it rests on this transparency and on the offline, no-bank-API design that avoids linking external accounts.

7. Linkage to Needs Assessment

Each FHS component directly addresses a confirmed user problem:

Needs Assessment Finding	FHS Component	Survey Question
Difficulty saving money	Savings Rate — rewards ≥20% savings	Q11: "Do you set aside money for savings?"
Regularly exceeds planned budget	Overspend Control — penalizes habitual overspend	Q14: "How often do expenses exceed budget?"
Fewer than half maintain any budget	Budget Adherence — incentivizes budgeting	Q8: "Do you prepare a budget?"
Inconsistent or no expense tracking	Logging Consistency — rewards daily logging	Q21: "Do you currently record expenses?"

8. Calibrated Constants (Design Choices)

The following numeric constants are design calibrations informed by the referenced frameworks but not directly derived from any single paper. They are disclosed as reasonable prototype values subject to empirical calibration in future longitudinal studies.

Constant	Value	Theoretical Basis	Design Rationale
Warning Decay rate	−5 pts/day	Prospect Theory (loss salience)	Felt but not catastrophic in a 30-day cycle
Warning Decay cap	3 days / −15 pts max	Prospect Theory	Prevents total wipe-out from single breach
Gap bonus	+2 pts/day (max +10)	Consistency principle	Smaller than penalty (loss-aversion asymmetry)
Gap penalty	−3 pts/day (max −15)	Prospect Theory asymmetry	Penalty > reward mirrors loss > gain finding
Soft overspend weight	0.5×	Nudge Theory (planned vs. habitual)	One-time large purchases ≠ daily excess
Category Balance threshold	40%	Concentration risk principle	Below 40%, dominance is minimal
Spending Restraint grace zone	80%	Personal finance buffer principle	20% safety buffer before penalty
Habit Streak target	14 days	SDT Competence + Lally et al. (2010)	Earliest reliable range for habit anchoring
Savings Rate target	20%	Warren & Tyagi 50/30/20 rule	Standard personal finance savings benchmark

9. References

All citations are verifiable via DOI, official publication page, or direct link.

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This document was prepared for adviser validation of the FHS formula derivation. All formulas are verified against the SmartSpend source code (`lib/services/score_service.dart`). All citations are verifiable.